

15.1.1: Car Dealership Sales System

ALGORITHM

1. START

2. DEFINE a base class Car:

- **METHOD** `__init__(self, brand, price, model, color)`:
 - Initialize the attributes `self.brand`, `self.price`, `self.model`, and `self.color`.

- **METHOD** `display_details(self)`:
 - Print the attributes of the car.

3. DEFINE derived classes Car1 and Car2 inheriting from Car:

- **OVERRIDE** the `display_details(self)` method in both classes to print car information (brand, price, model, and color) each on a separate line.

4. MAIN PROGRAM:

- **READ** the input for Car1 (brand, price, model, color) from a single line.
- **READ** the input for Car2 (brand, price, model, color) from a second line.
- **INstantiate** objects for Car1 and Car2 with the provided inputs.
- **CALL** the `display_details()` method for the Car1 instance.
- **CALL** the `display_details()` method for the Car2 instance.

5. END

15.1.1. Car Dealership Sales System

Write a Python program to model a car dealership sales system using object-oriented programming. Create a base class `Car` and two derived classes `Car1` and `Car2` to represent different car types.

Class Structure

- Base Class `Car`:
 - Attributes: `brand`, `price`, `model`, `color`.
 - Method: `display_details()` to print all car attributes.
- Derived Classes `Car1` and `Car2`:
 - Inherit from `Car` class.
 - Override `display_details()` method to display car information in the specified format.

Input Format:

- The first line contains `Car1`'s `brand` (string), `price` (float), `model` (string), and `color` (string) separated by spaces.
- The second line contains `Car2`'s `brand` (string), `price` (float), `model` (string), and `color` (string) separated by spaces.

Output Format:

- First four lines: `Car1` details (`brand`, `price`, `model`, `color` - each on separate line)
- Next four lines: `Car2` details (`brand`, `price`, `model`, `color` - each on separate line)

Sample Test Cases

```
1 # Base class
2 class Car:
3     def __init__(self, brand, price, model, color):
4         self.brand = brand
5         self.price = float(price)
6         self.model = model
7         self.color = color
8
9     def display_details(self):
10        print(self.brand)
11        print(self.price)
12        print(self.model)
13        print(self.color)
```

Test Results:

- Average time: 0.011 s (11.00 ms)
- Maximum time: 0.021 s (21.00 ms)
- 3 out of 3 shown test case(s) passed
- 3 out of 3 hidden test case(s) passed

Expected output	Actual output
Toyota 25000.50 Camry White	Toyota 25000.50 Camry White
Honda 22000.00 Accord Black	Honda 22000.00 Accord Black
Toyota:	Toyota:
25000.5	25000.5
Camry	Camry
White	White

